Standalone simulations (Glass Simulations)

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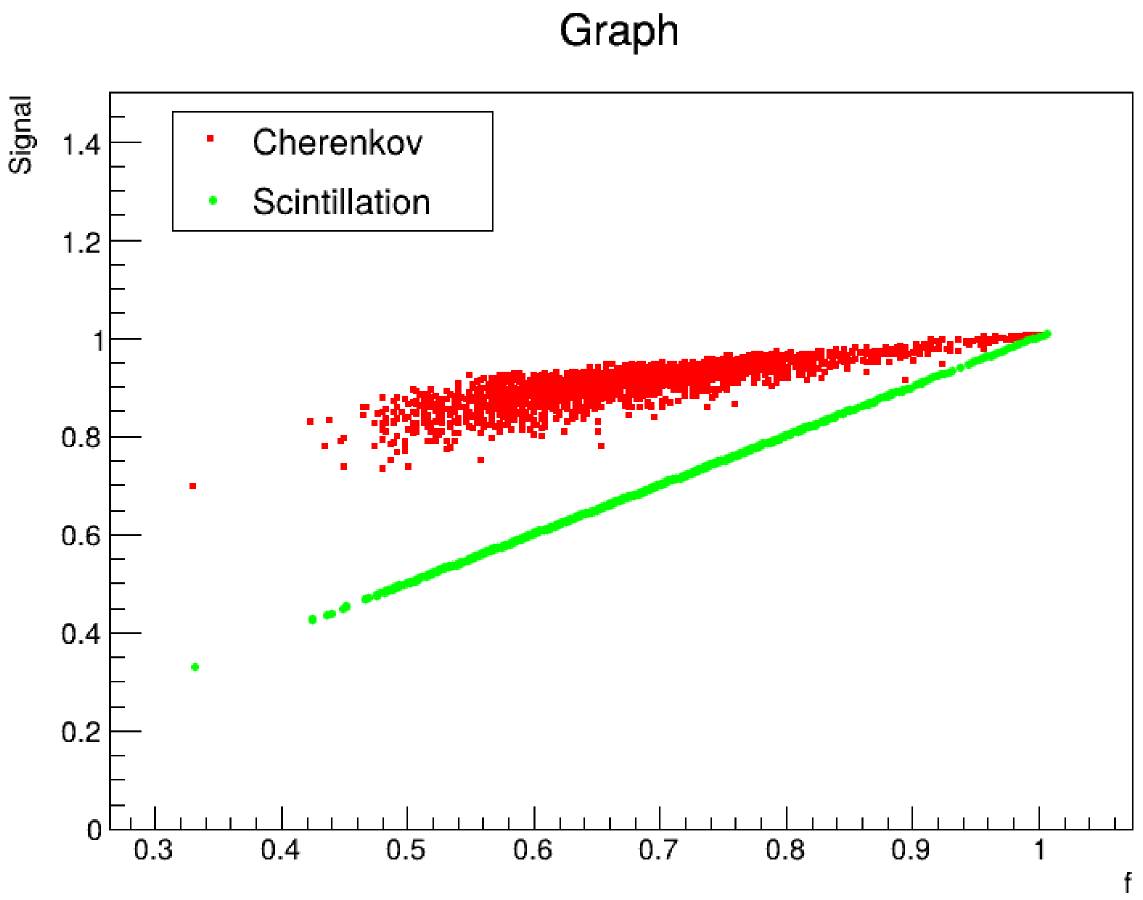
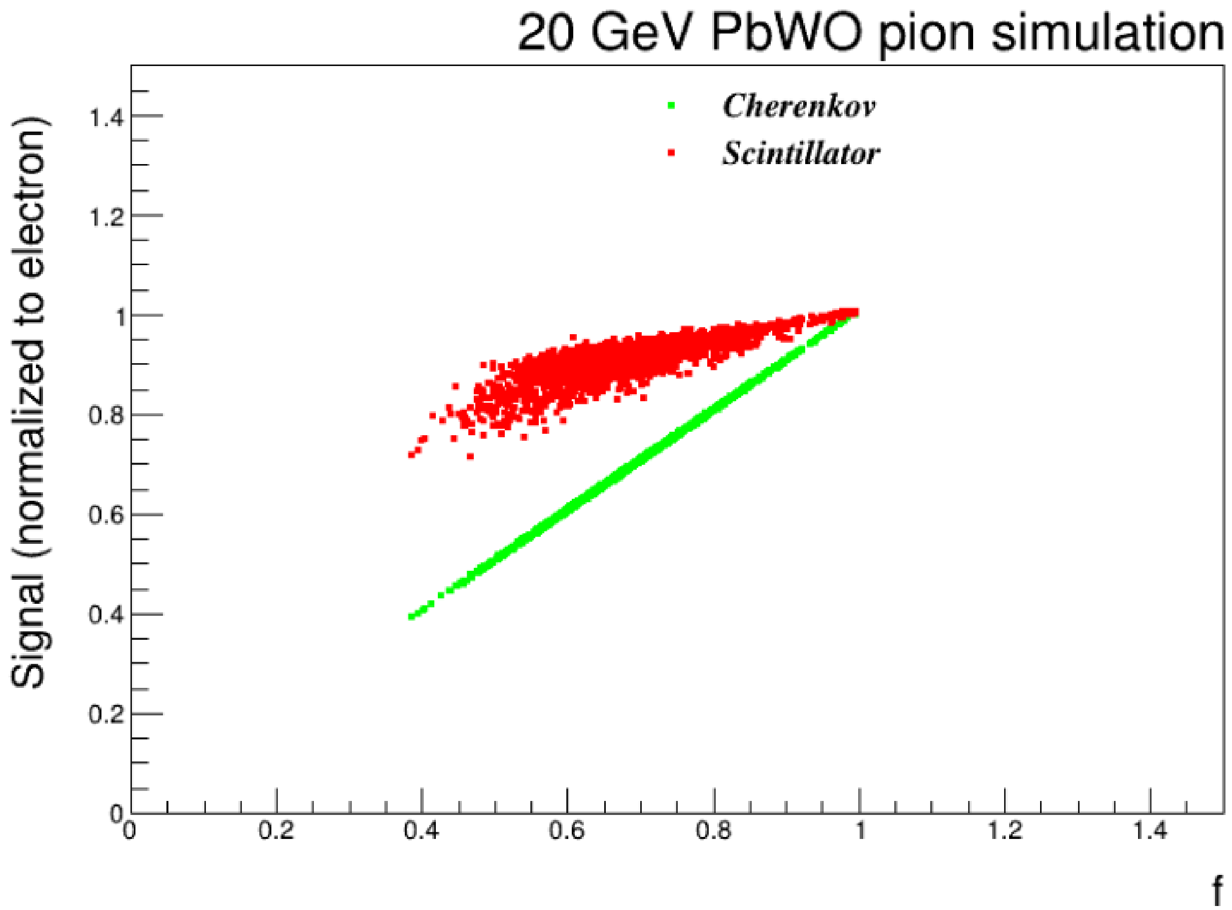
Simulations studies for Glass

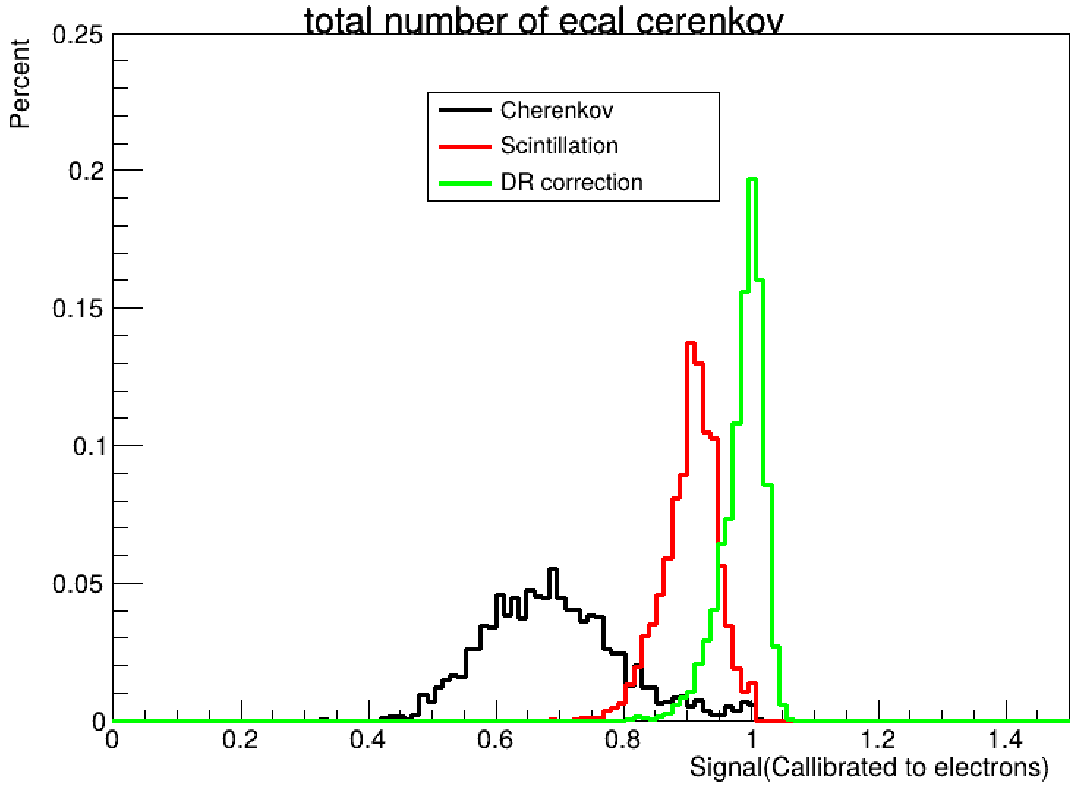
- Tried reproducing the simulation plots for PbWO4 and compared with the ones in arxiv: 2501.15329v3.
- The simulations were generated for 2000 events.
- We also generated those plots for a template Glass(ABS) description for ECAL, replacing PbWO4.
    - Glass_ABS is defined for Glass Scintillator.
    - Scaled down the Refractive index for ABS using PBWO4 R-index.
    - Rest of the parameters are kept similar to PBWO4

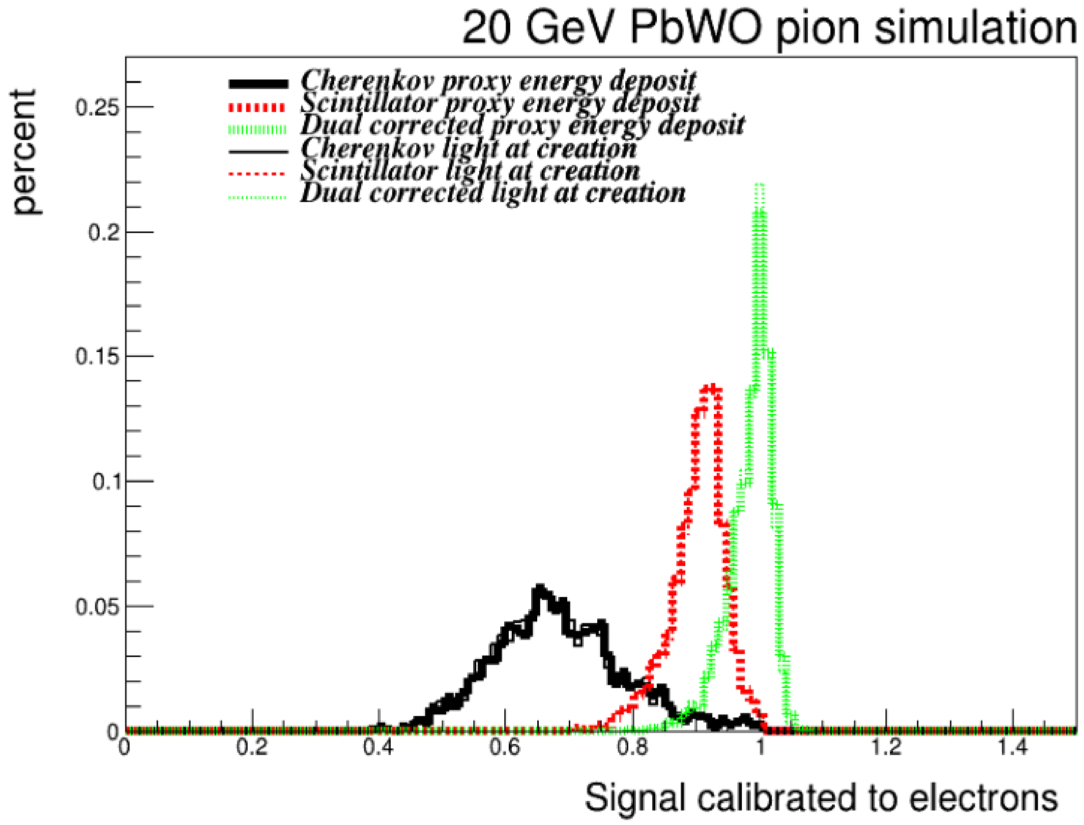
Simulations vs 2501.15329v3



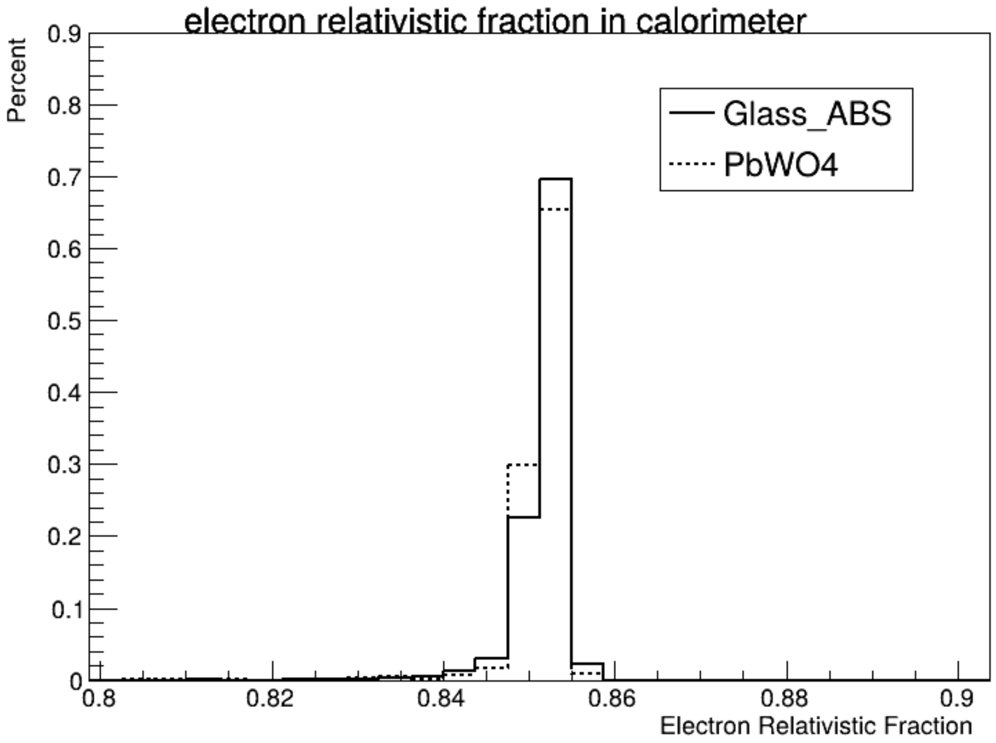


Simulations vs 2501.15329v3



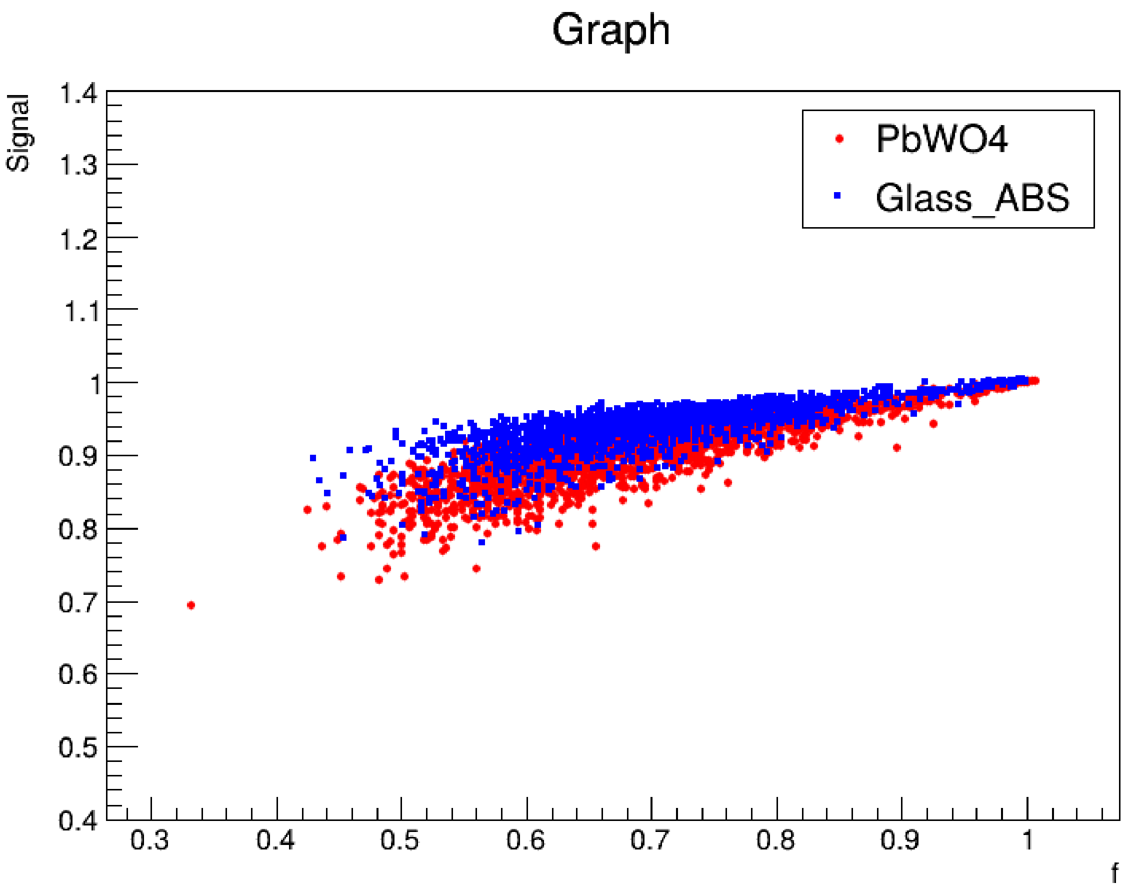


Glass vs PbWO4(2000 Events)

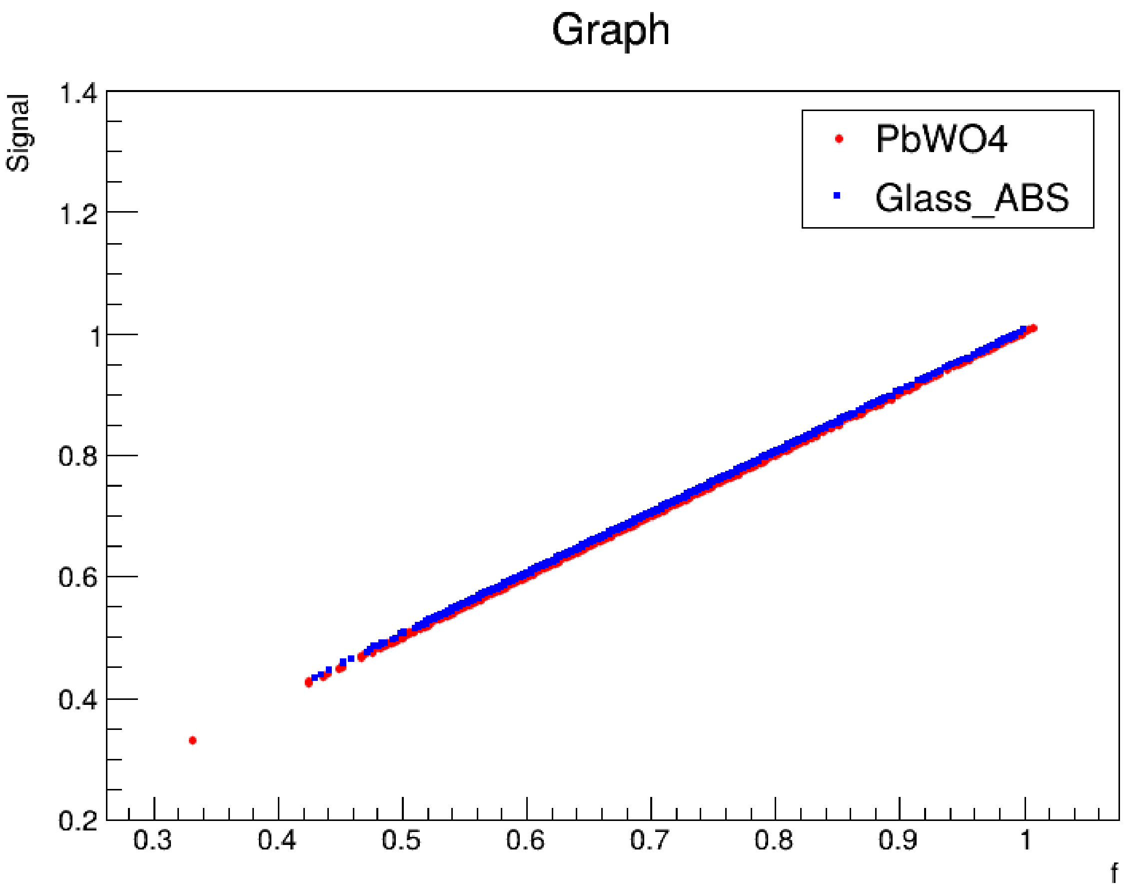


•Fraction of deposited ionizing energy in 20 GeV electron showers deposited by particles Fraction of deposited ionizing energy in 20 GeV electron showers deposited by particles

Glass vs PbWO4(2000 Events)

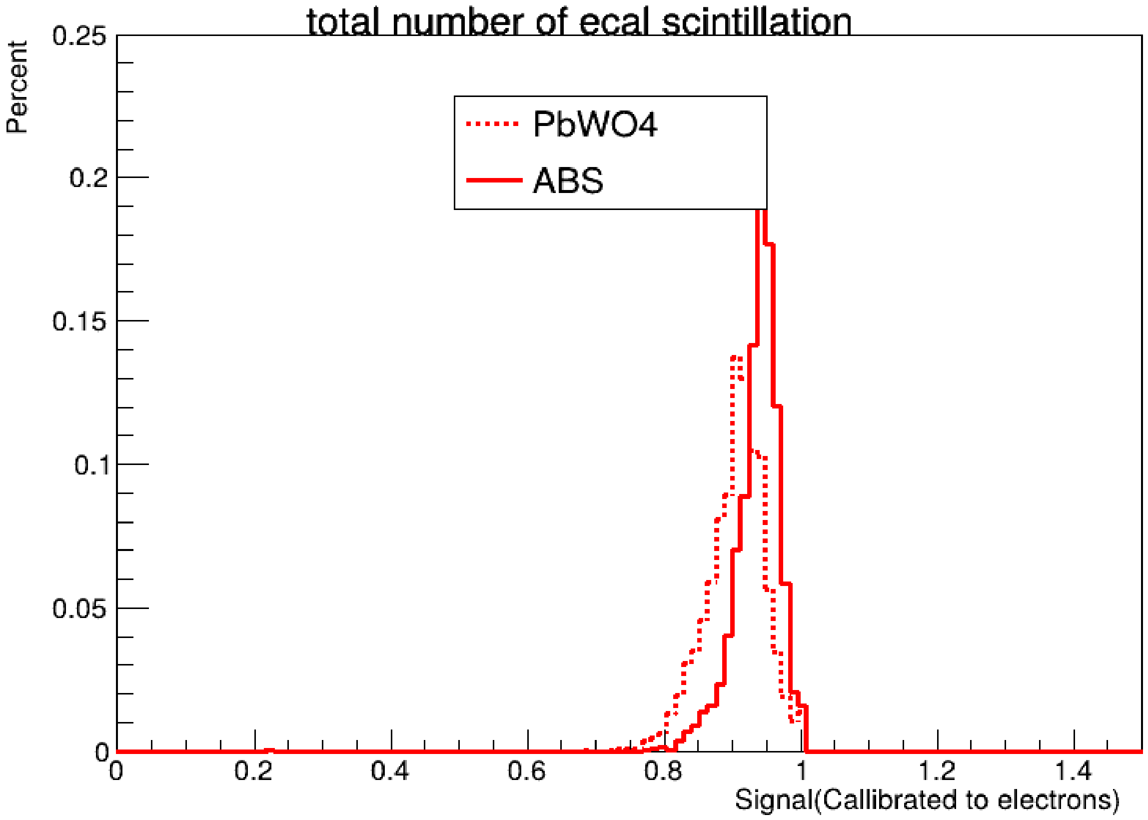


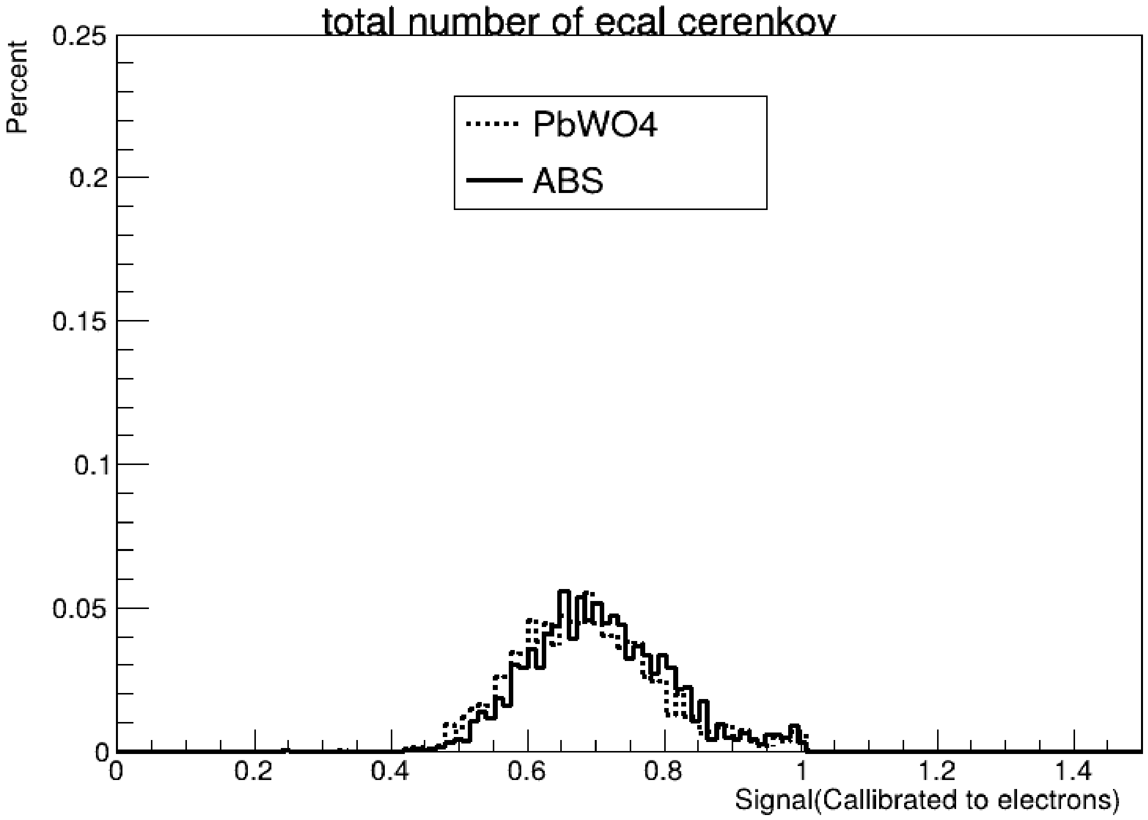
Scintillation Signal



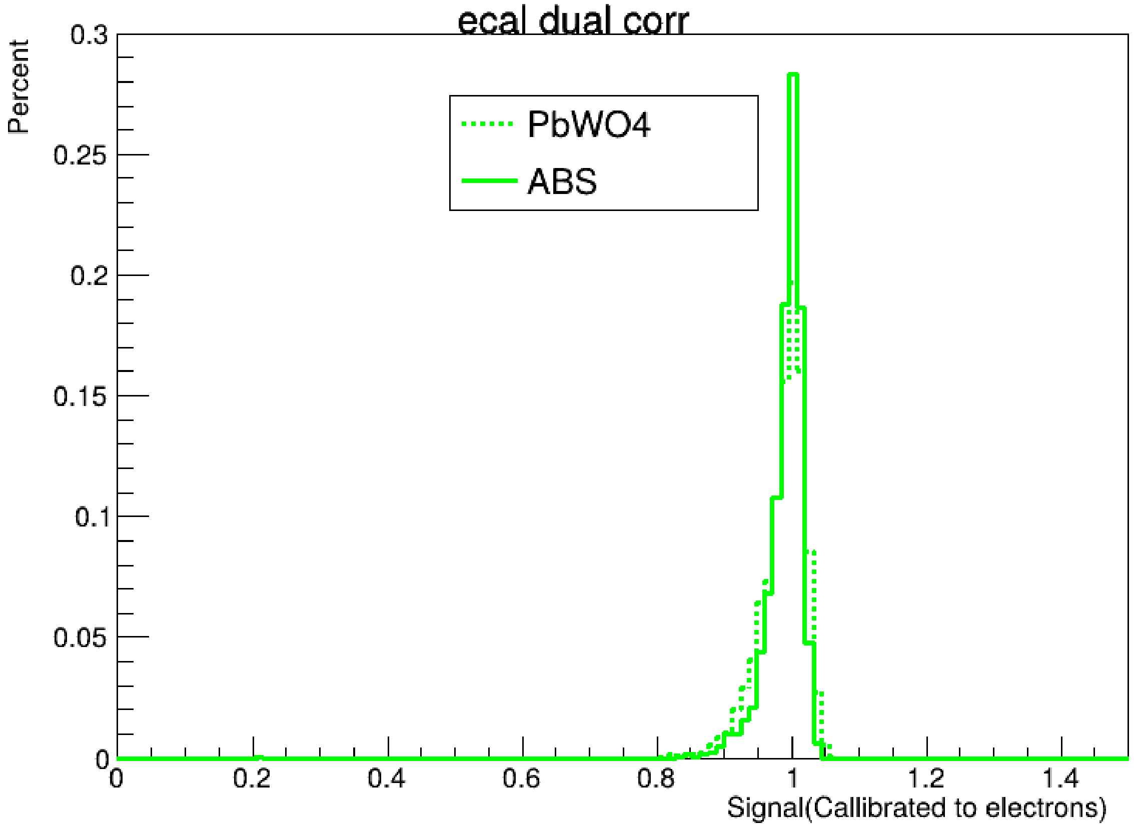
Cherenkov Signal

Glass vs PbWO4(2000 Events)

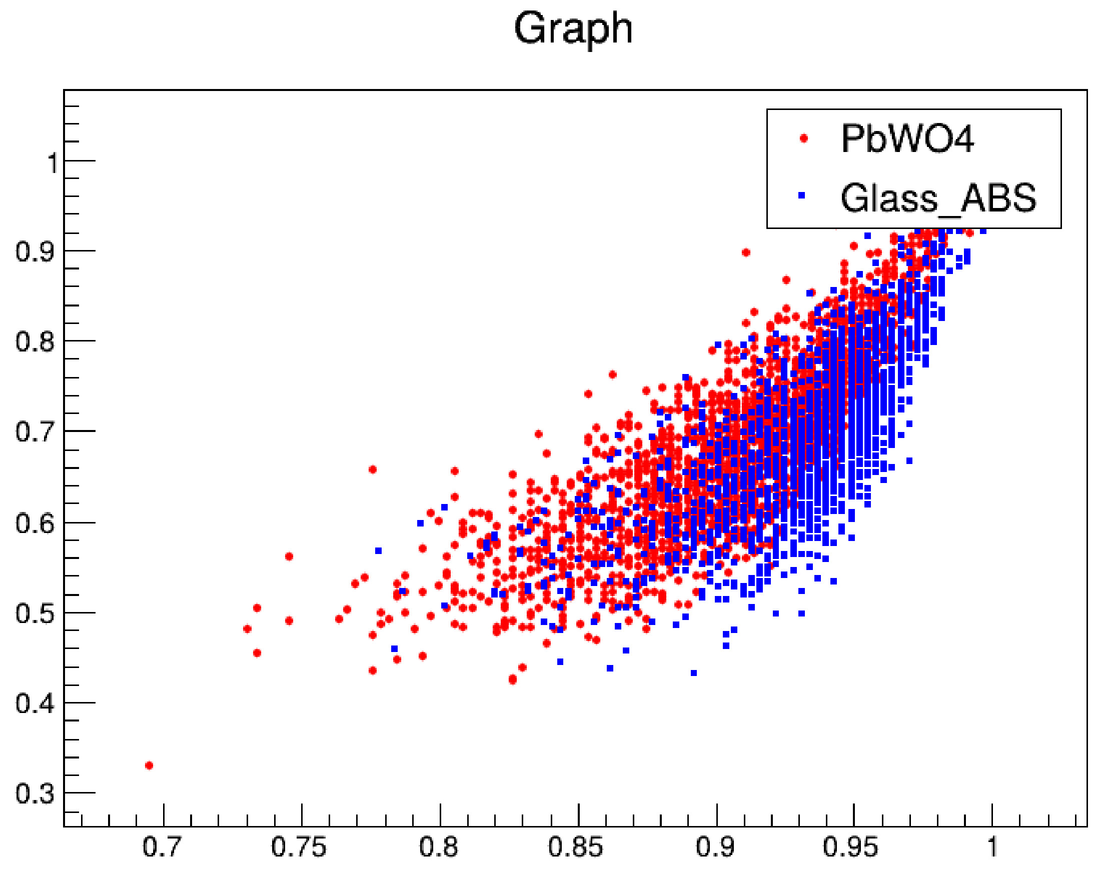




Glass vs PbWO4(2000 Events)



Dual Readout Correction



Cherenkov versus scintillation signal for calorimeters